電子學實驗一
Integrators, Differentiators, and Their Frequency Responses
預報

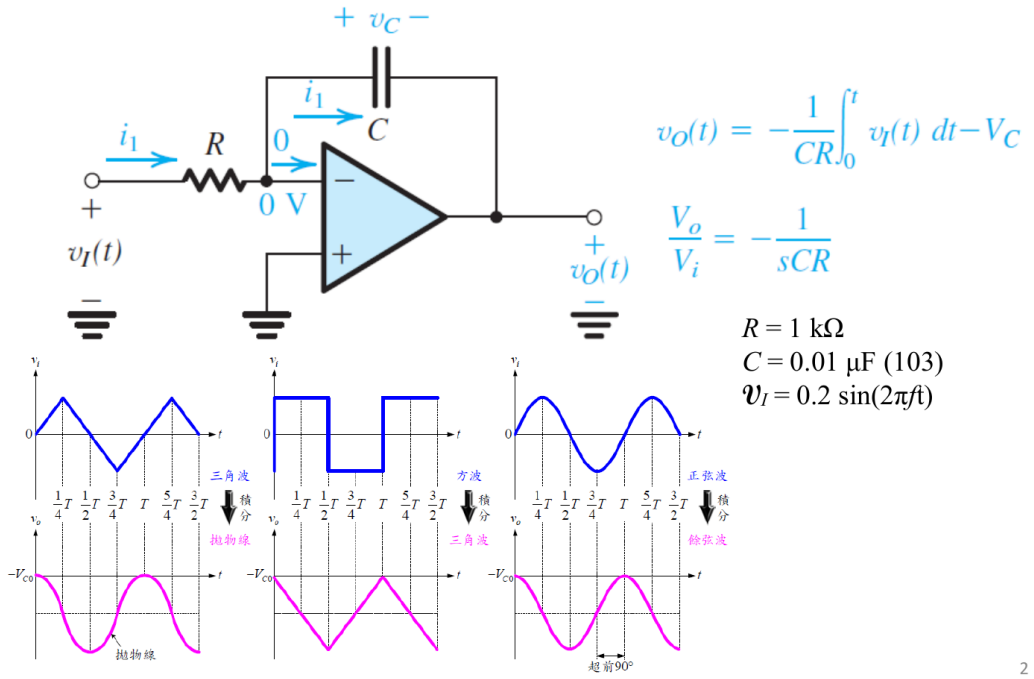
B14901078 林孜謙、B14901153 沈威宇

1 實驗目的

- To understand the operating principles of Miller integrator and its feasible configuration with feedback resistor in parallel connection, and differentiator circuits.
- To observe their output waveform under triangular, square, and sine waves.
- To measure their frequency responses and sketch the Bode magnitude plots.

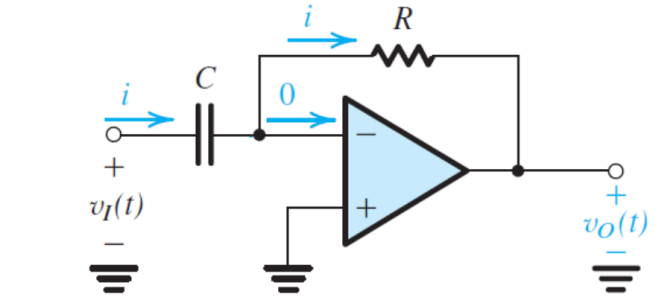
2 預備實驗的電路圖

Exp. 1 Integrator



2

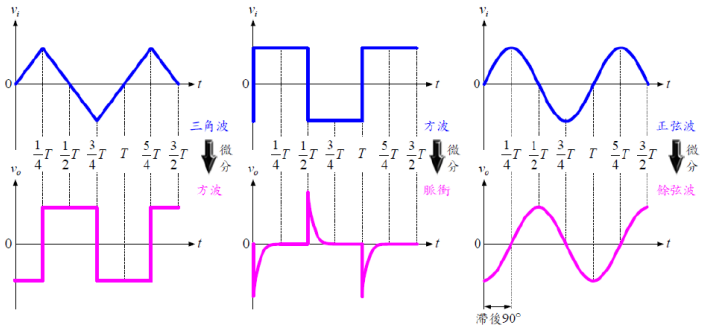
Exp. 2 Differentiator



$$i(t) = C\frac{dv_I(t)}{dt}$$

$$v_O(t) = -CR\frac{dv_I(t)}{dt}$$

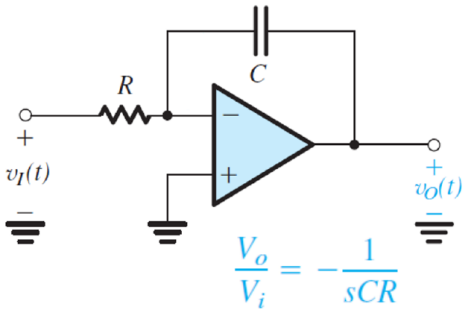
$$\frac{V_o}{V_i} = -sCR$$



$R = 1$ kΩ
$C = 0.01$ μF (103)
$v_I = 0.2\sin(2\pi ft)$

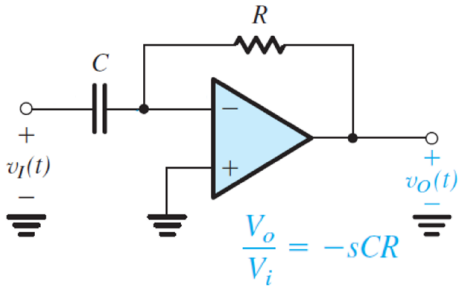
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Exp. 3 Frequency Response (I)



$$\frac{V_o}{V_i} = -\frac{1}{sCR}$$

$R = 1$ kΩ
$C = 0.01$ μF (103)
$v_I = 0.2\sin(2\pi ft)$, $f = 100$ Hz ~ 300 kHz
Please sketch the Bode magnitude plot.

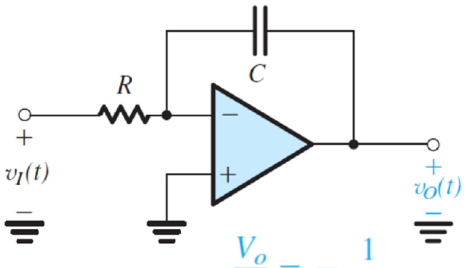


$$\frac{V_o}{V_i} = -sCR$$

$R = 1$ kΩ
$C = 0.01$ μF (103)
$v_I = 0.2\sin(2\pi ft)$, $f = 100$ Hz ~ 100 kHz
Please sketch the Bode magnitude plot.

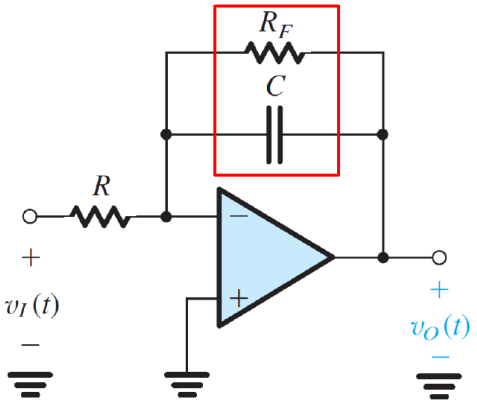
4

Exp. 3 Frequency Response (II)



$$\frac{V_o}{V_i} = -\frac{1}{sCR}$$

$R = 1\ \text{k}\Omega$
$C = 0.001\ \mu\text{F}$ (102)
$v_I = 0.2\sin(2\pi ft), f = 100\ \text{Hz} \sim 1\ \text{MHz}$
Please sketch the Bode magnitude plot.



$R = 1\ \text{k}\Omega$, $R_F = ?\ \text{k}\Omega$
$C = 0.001\ \mu\text{F}$ (102)
$v_I = 0.2\sin(2\pi ft), f = 100\ \text{Hz} \sim 1\ \text{MHz}$
Please sketch the Bode magnitude plot.

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3 預備量測的數據